- **Standardize Features:** Ensure all features contribute equally by scaling them to have a mean of 0 and a standard deviation of 1.

```
from sklearn.preprocessing import StandardScaler scaler = StandardScaler()
df_scaled = scaler.fit_transform(df[['feature1', 'feature2']])
```

Handling Outliers

To minimize the impact of outliers, consider removing them or using robust versions of K-Means, such as MiniBatchKMeans or algorithms based on density clustering.

Insights from K-Means

1. **Cluster Assignments:** K-Means provides cluster labels for each data point, enabling segmentation and deeper analysis.
2. **Centroid Locations:** The centroids represent the "average" characteristics of each cluster, offering a summary of the data.
3. **Visualization:** Visual tools such as scatter plots can help interpret clustering results and identify patterns within the data.

K-Means is a powerful and efficient clustering algorithm that excels in identifying groupings within datasets. Despite its limitations, it remains a cornerstone of unsupervised learning due to its simplicity and interpretability. By carefully preprocessing data, selecting the appropriate value for k, and evaluating cluster quality, K-Means can provide valuable insights and drive informed decision-making across various domains.

Dimensionality Reduction: PCA Implementation

High-dimensional datasets often pose challenges in analysis and visualization, as they may contain redundant or irrelevant features that complicate models and increase computational costs. Dimensionality reduction techniques like **Principal Component Analysis (PCA)** help address these issues by transforming data into a lower-dimensional space while preserving as much variance as possible. PCA is a cornerstone method in data preprocessing and exploratory data analysis, offering insights into the structure of high-dimensional datasets.

How PCA Works